EXPLANATION_EVIDENCE

uuid	explanation_id	PK,FK
uuid	evidence_item_id	PK,FK
varchar	relationship_label	

EVIDENCE_ITEMS

uuid	evidence_item_id	PK
uuid	data_source_id	FK
uuid	security_id	FK
uuid	market_price_id	FK
uuid	fundamental_metric_id	FK
uuid	economic_observation_id	FK
uuid	news_article_id	FK
varchar	evidence_type	
varchar	title	
varchar	evidence_summary	
varchar	source_url	
varchar	source_record_reference	
timestamp	published_at	
timestamp	observed_at	
jsonb	cited_value_snapshot	
timestamp	created_at	

PORTFOLIO_VALIDATION_RESULTS

uuid	validation_result_id	PK
uuid	validation_run_id	FK
varchar	rule_code	
varchar	result_status	
varchar	severity	
uuid	security_id	FK
numeric	actual_value	
numeric	threshold_value	
varchar	result_message	
jsonb	details	
timestamp	created_at	

RECOMMENDATION_EXPLANATIONS

uuid	explanation_id	PK
uuid	portfolio_version_id	FK
uuid	decision_id	FK
varchar	explanation_type	
varchar	title	
varchar	explanation_text	
integer	display_order	
timestamp	created_at	

PORTFOLIO_SCENARIOS

uuid	portfolio_scenario_id	PK
uuid	portfolio_version_id	FK
varchar	scenario_code	
varchar	scenario_name	
varchar	scenario_type	
jsonb	assumptions	
numeric	estimated_portfolio_value	
numeric	estimated_change_amount	
numeric	estimated_change_pct	
varchar	result_summary	
varchar	calculation_version	
timestamp	created_at	

PORTFOLIO_SECURITY_DECISIONS

uuid	decision_id	PK
uuid	portfolio_version_id	FK
uuid	security_id	FK
uuid	listing_id	FK
varchar	decision_status	
numeric	target_weight_pct	
varchar	decision_reason	
timestamp	created_at	

PORTFOLIO_VALIDATION_RUNS

uuid	validation_run_id	PK
uuid	portfolio_version_id	FK
varchar	rule_set_version	
varchar	calculation_engine_version	
varchar	validation_status	
timestamp	started_at	
timestamp	completed_at	
jsonb	summary	

PORTFOLIO_VERSION_CONSTRAINTS

uuid	portfolio_constraint_id	PK
uuid	portfolio_version_id	FK
varchar	constraint_code	
varchar	operator	
numeric	comparison_value	
numeric	upper_value	
varchar	unit_code	
boolean	is_mandatory	
boolean	applied	
varchar	description	
timestamp	created_at	

PORTFOLIO_VERSIONS

uuid	portfolio_version_id	PK
uuid	portfolio_id	FK
uuid	generation_request_id	FK
uuid	strategy_version_id	FK
integer	version_number	
varchar	version_status	
numeric	investment_amount	
varchar	currency_code	
integer	investment_horizon_months	
varchar	risk_category	
numeric	expected_return_pct	
numeric	expected_volatility_pct	
numeric	diversification_score	
timestamp	market_context_as_of	
jsonb	generation_context	
timestamp	generated_at	
timestamp	created_at	

PORTFOLIO_GENERATION_REQUESTS

uuid	generation_request_id	PK
uuid	portfolio_id	FK
uuid	user_id	FK
uuid	goal_id	FK
uuid	risk_assessment_id	FK
varchar	request_type	
varchar	request_status	
varchar	user_prompt	
numeric	requested_amount	
varchar	currency_code	
integer	investment_horizon_months	
jsonb	normalized_inputs	
jsonb	investor_snapshot	
timestamp	market_context_as_of	
timestamp	started_at	
timestamp	completed_at	
varchar	error_code	
varchar	error_summary	
timestamp	created_at	

STRATEGY_VERSIONS

uuid	strategy_version_id	PK
uuid	strategy_id	FK
varchar	version_label	
varchar	optimizer_name	
varchar	implementation_revision	
jsonb	configuration	
varchar	constraint_set_version	
timestamp	created_at	

INVESTOR_PORTFOLIOS

uuid	portfolio_id	PK
uuid	user_id	FK
varchar	portfolio_name	
varchar	portfolio_status	
timestamp	created_at	
timestamp	updated_at	

STRATEGY_DEFINITIONS

uuid	strategy_id	PK
varchar	strategy_code	
varchar	strategy_name	
varchar	description	
boolean	is_active	
timestamp	created_at	